

UNIVERSITY OF ENGINEERING & TECHNOLOGY, MARDAN

Department of Computer Science

DSA LAB REPORT

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Section	A
Semester	3rd
Course	Data Structures & Algorithms (DSA)
Lab No.	01

Part A: Compiler Installation

```

Dev@DESKTOP-I0VAMRS UCRT64 ~
$ $ pacman -S mingw-w64-ucrt-x86_64-gcc
-bash: $: command not found

Dev@DESKTOP-I0VAMRS UCRT64 ~
$ pacman -S mingw-w64-ucrt-x86_64-gcc
resolving dependencies...
looking for conflicting packages...

Packages (17) mingw-w64-ucrt-x86_64-binutils-2.46-4
               mingw-w64-ucrt-x86_64-crt-14.0.0.r92.g818fa6510-1
               mingw-w64-ucrt-x86_64-gcc-libs-16.1.0-5  mingw-w64-ucrt-x86_64-gettext-runtime-1.0-1
               mingw-w64-ucrt-x86_64-gmp-6.3.0-2
               mingw-w64-ucrt-x86_64-headers-14.0.0.r92.g818fa6510-1
               mingw-w64-ucrt-x86_64-isl-0.27-1  mingw-w64-ucrt-x86_64-libconv-1.19-1
               mingw-w64-ucrt-x86_64-libwinpthread-14.0.0.r92.g818fa6510-1
               mingw-w64-ucrt-x86_64-mpc-1.4.1-1  mingw-w64-ucrt-x86_64-mpfr-4.2.2-3
               mingw-w64-ucrt-x86_64-tzdata-2026b-1
               mingw-w64-ucrt-x86_64-windows-default-manifest-6.4-4
               mingw-w64-ucrt-x86_64-winpthread-14.0.0.r92.g818fa6510-1
               mingw-w64-ucrt-x86_64-zlib-1.3.2-2  mingw-w64-ucrt-x86_64-zstd-1.5.7-2
               mingw-w64-ucrt-x86_64-gcc-16.1.0-5

Total Download Size: 68.98 MiB
Total Installed Size: 490.23 MiB

:: Proceed with installation? [Y/n] y
:: Retrieving packages...
mingw-w64-ucrt-x86_64-isl... 1453.5 KiB 274 KiB/s 00:05 [#####] 100%
mingw-w64-ucrt-x86_64-gcc-libs-16.1.0-5-any 1125.3 KiB 156 KiB/s 00:07 [#####] 100%
mingw-w64-ucrt-x86_64-binutils-2.46-4-any 879.9 KiB 109 KiB/s 00:50 [#####] 13%
mingw-w64-ucrt-x86_64-crt-14.0.0.r92.g818fa6510-1 207.8 KiB 9.86 KiB/s 07:41 [#####] 4%
mingw-w64-ucrt-x86_64-gcc-16.1.0-5-any 1503.8 KiB 146 KiB/s 05:09 [#####] 3%
mingw-w64-ucrt-x86_64-headers-14.0.0.r92.g818f... 5.1 MiB 594 KiB/s 00:02 [#####] 77%
mingw-w64-ucrt-x86_64-libconv-1.19-1-any 9.4 MiB 640 KiB/s 01:35 [#####] 13%
Total ( 2/17) 10.1 MiB 871 KiB/s 01:09 [#####] 14%

(15/38) installing mingw-w64-ucrt-x86_64-wineditline [#####] 100%
(16/38) installing mingw-w64-ucrt-x86_64-pcre2 [#####] 100%
(17/38) installing mingw-w64-ucrt-x86_64-libtne [#####] 100%
(18/38) installing mingw-w64-ucrt-x86_64-libsystre [#####] 100%
(19/38) installing mingw-w64-ucrt-x86_64-ncurses [#####] 100%
(20/38) installing mingw-w64-ucrt-x86_64-libffi [#####] 100%
(21/38) installing mingw-w64-ucrt-x86_64-mpdecimal [#####] 100%
(22/38) installing mingw-w64-ucrt-x86_64-openssl [#####] 100%
Optional dependencies for mingw-w64-ucrt-x86_64-openssl
mingw-w64-ucrt-x86_64-ca-certificates
(23/38) installing mingw-w64-ucrt-x86_64-tcl [#####] 100%
(24/38) installing mingw-w64-ucrt-x86_64-sqlite3 [#####] 100%
Optional dependencies for mingw-w64-ucrt-x86_64-sqlite3
mingw-w64-ucrt-x86_64-tcl: for sqlite3_analyzer [installed]
(25/38) installing mingw-w64-ucrt-x86_64-tk [#####] 100%
(26/38) installing mingw-w64-ucrt-x86_64-libb2 [#####] 100%
(27/38) installing mingw-w64-ucrt-x86_64-xz [#####] 100%
(28/38) installing mingw-w64-ucrt-x86_64-python [#####] 100%
(29/38) installing mingw-w64-ucrt-x86_64-termcap [#####] 100%
(30/38) installing mingw-w64-ucrt-x86_64-readline [#####] 100%
(31/38) installing mingw-w64-ucrt-x86_64-xxhash [#####] 100%
(32/38) installing mingw-w64-ucrt-x86_64-gdb [#####] 100%
Optional dependencies for mingw-w64-ucrt-x86_64-gdb
mingw-w64-ucrt-x86_64-python-pygments: for syntax highlighting
(33/38) installing mingw-w64-ucrt-x86_64-gdb-multiarch [#####] 100%
Optional dependencies for mingw-w64-ucrt-x86_64-gdb-multiarch
mingw-w64-ucrt-x86_64-python-pygments: for syntax highlighting
(34/38) installing mingw-w64-ucrt-x86_64-libmangle [#####] 100%
(35/38) installing mingw-w64-ucrt-x86_64-make [#####] 100%
(36/38) installing mingw-w64-ucrt-x86_64-pkgconf [#####] 100%
(37/38) installing mingw-w64-ucrt-x86_64-tools [#####] 100%
(38/38) installing mingw-w64-ucrt-x86_64-winstorescompat [#####] 100%
:: Running post-transaction hooks...
(1/1) Updating the info directory file...

Dev@DESKTOP-I0VAMRS UCRT64 ~
$

```

Task 01: Hello DSA

Output



The screenshot shows the Visual Studio Code editor with a C++ file named `Hello_DSA.cc` open. The code is as follows:

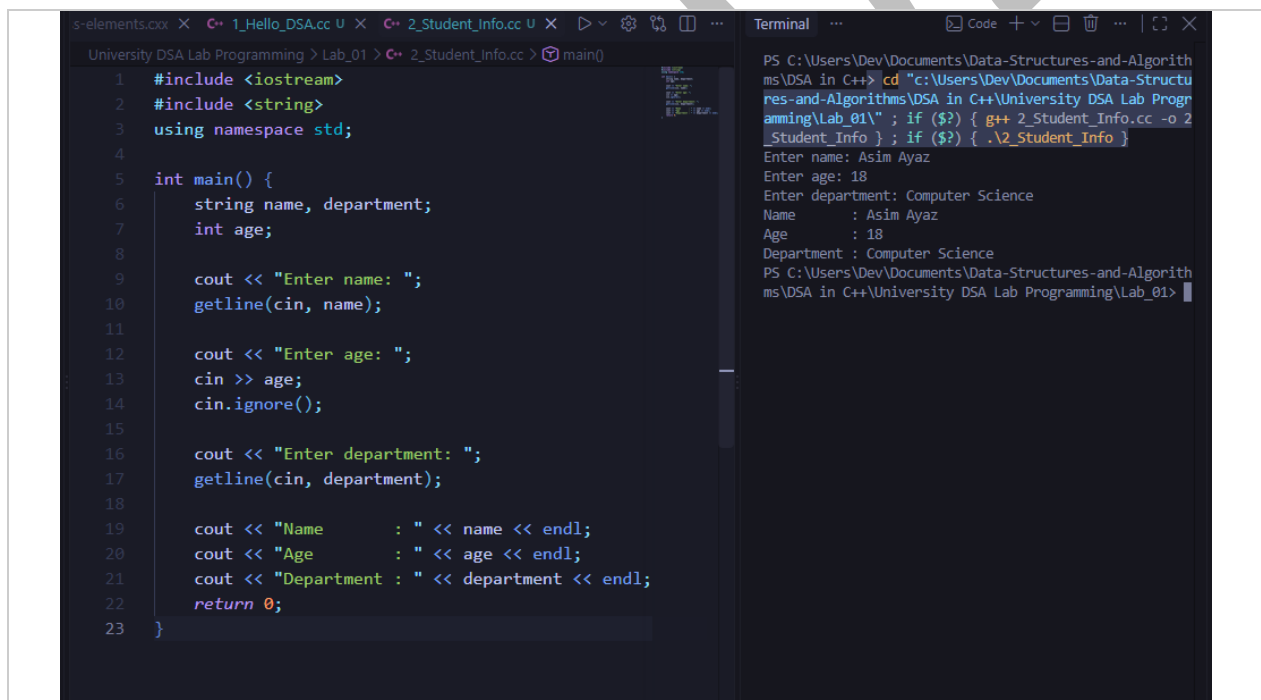
```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     cout << "Hello DSA!";
6     return 0;
7 }
```

The terminal output shows the command to compile and run the program, and the resulting output:

```
PS C:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++\University DSA Lab Programming\Lab_01> cd "C:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++\University DSA Lab Programming\Lab_01" ; if ($?) { g++ Hello_DSA.cc -o Hello_DSA } ; if ($?) { .\Hello_DSA }
Hello DSA!
PS C:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++\University DSA Lab Programming\Lab_01>
```

Task 02: Student Information

Output



The screenshot shows the Visual Studio Code editor with a C++ file named `2_Student_Info.cc` open. The code is as follows:

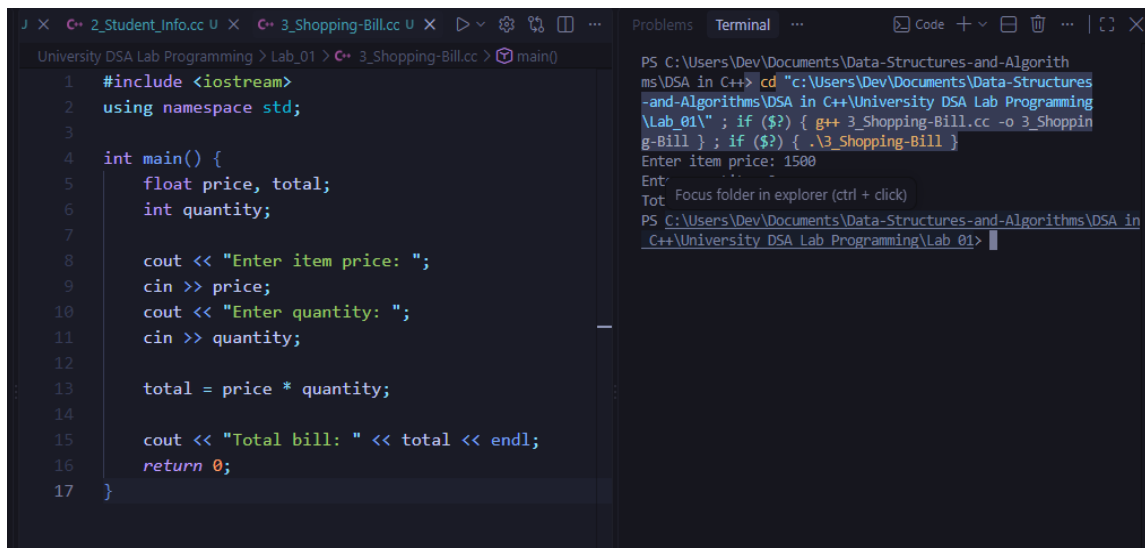
```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 int main() {
6     string name, department;
7     int age;
8
9     cout << "Enter name: ";
10    getline(cin, name);
11
12    cout << "Enter age: ";
13    cin >> age;
14    cin.ignore();
15
16    cout << "Enter department: ";
17    getline(cin, department);
18
19    cout << "Name      : " << name << endl;
20    cout << "Age       : " << age << endl;
21    cout << "Department : " << department << endl;
22    return 0;
23 }
```

The terminal output shows the command to compile and run the program, and the resulting output:

```
PS C:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++\University DSA Lab Programming\Lab_01> cd "C:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++\University DSA Lab Programming\Lab_01" ; if ($?) { g++ 2_Student_Info.cc -o 2_Student_Info } ; if ($?) { .\2_Student_Info }
Enter name: Asim Ayaz
Enter age: 18
Enter department: Computer Science
Name      : Asim Ayaz
Age       : 18
Department : Computer Science
PS C:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++\University DSA Lab Programming\Lab_01>
```

Task 03: Shopping Bill Calculator

Output

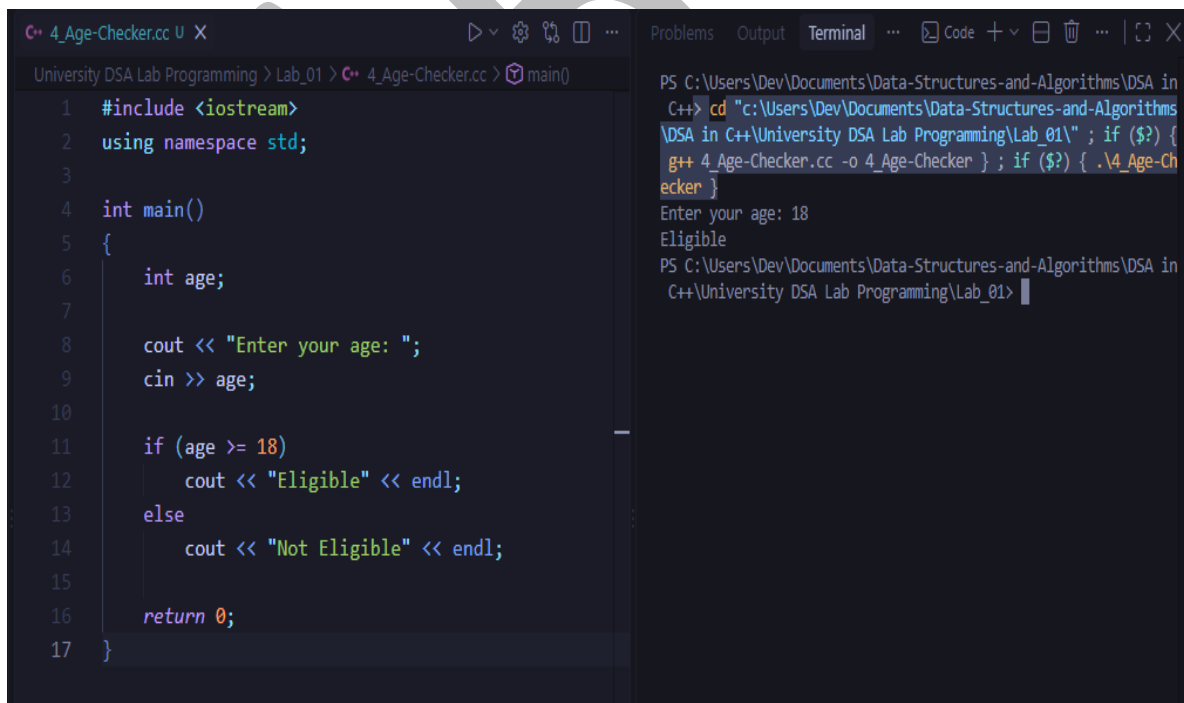


```
University DSA Lab Programming > Lab_01 > C++ 3_Shopping-Bill.cc > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      float price, total;
6      int quantity;
7
8      cout << "Enter item price: ";
9      cin >> price;
10     cout << "Enter quantity: ";
11     cin >> quantity;
12
13     total = price * quantity;
14
15     cout << "Total bill: " << total << endl;
16     return 0;
17 }
```

```
PS C:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++> cd "c:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++\University DSA Lab Programming\Lab_01\"; if ($?) { g++ 3_Shopping-Bill.cc -o 3_Shopping-Bill }; if ($?) { .\3_Shopping-Bill }
Enter item price: 1500
Enter quantity: 10
Total bill: 15000
PS C:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++\University DSA Lab Programming\Lab_01>
```

Task 04: Age Eligibility Checker

Output



```
C++ 4_Age-Checker.cc U X
University DSA Lab Programming > Lab_01 > C++ 4_Age-Checker.cc > main()
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int age;
7
8      cout << "Enter your age: ";
9      cin >> age;
10
11     if (age >= 18)
12     {
13         cout << "Eligible" << endl;
14     }
15     else
16     {
17         cout << "Not Eligible" << endl;
18     }
19     return 0;
20 }
```

```
PS C:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++> cd "c:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++\University DSA Lab Programming\Lab_01\"; if ($?) { g++ 4_Age-Checker.cc -o 4_Age-Checker }; if ($?) { .\4_Age-Checker }
Enter your age: 18
Eligible
PS C:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++\University DSA Lab Programming\Lab_01>
```

Task 05: Even or Odd

Output

```
C++ 4_Age-Checker.cc U X C++ 5_Even-or-Odd.cc U X
University DSA Lab Programming > Lab_01 > C++ 5_Even-or-Odd.cc > main()
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int number;
6
7     cout << "Enter a number: ";
8     cin >> number;
9
10    if (number % 2 == 0)
11        cout << number << " is Even" << endl;
12    else
13        cout << number << " is Odd" << endl;
14
15    return 0;
16 }
```

```
PS C:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++> cd "c:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++\University DSA Lab Programming\Lab_01\" ; if ($?) { g++ 5_Even-or-Odd.cc -o 5_Even-or-Odd } ; if ($?) { .\5_Even-or-Odd }
Enter a number: 13
13 is Odd
PS C:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++\University DSA Lab Programming\Lab_01>
```

Task 06: Temperature Conversion

Output

```
C++ 4_Age-Checker.cc U X C++ 5_Even-or-Odd.cc U X C++ 6_Temperature-Conversion.cc U X
University DSA Lab Programming > Lab_01 > C++ 6_Temperature-Conversion.cc > main()
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     float celsius, fahrenheit;
6
7     cout<<"Enter temperature in Celsius: ";
8     cin>>celsius;
9
10    fahrenheit = (celsius * 9.0 / 5.0) + 32;
11
12    cout<<"Temperature in Fahrenheit: " <<fahrenheit<<endl;
13    return 0;
14 }
```

```
PS C:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++> cd "c:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++\University DSA Lab Programming\Lab_01\" ; if ($?) { g++ 6_Temperature-Conversion.cc -o 6_Temperature-Conversion } ; if ($?) { .\6_Temperature-Conversion }
Enter temperature in Celsius: 37.9
Temperature in Fahrenheit: 100.22
PS C:\Users\Dev\Documents\Data-Structures-and-Algorithms\DSA in C++\University DSA Lab Programming\Lab_01>
```